



Defence Innovation Portal (IDEaS and NATO DIANA)

Component 1a

Proposal

A proposal describes a proposed solution aimed at establishing the knowledge and technological advantage needed to address Canada's defence and security interests for the selected S&T Challenge.

The responses provided by the Offeror regarding the submission of the Technical and the Financial Proposal will form the Offeror proposal in its entirety.

Note: The following acronyms are used throughout the proposal and are defined as:

SC = Screening Criteria; **MC** = Mandatory Criteria; **PRC** = Point-Rated Criteria

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Company representative name: Steven Ness

Proposal Reference Number: CP6-144220

Category of Submission: Replacement Submission

Proposal reference number being replaced:: CP6-143774

Proposal Submitted Date: 2026-07-20 08:15 PM

Proposal Title

Fusion Compliance Engine: Sovereign Real-Time Classification Enforcement for Multi-Sensor Fusion

1 - Science and Technology (S&T) Challenge

A. S&T Challenge of the Proposed Solution

Identify the S&T Challenge being addressed by this proposal.

S&T Challenge of the Proposed Solution

Reliable AI Sensor Fusion for Real-World Missions

2 - Project Description

A. Project Synopsis

Provide a brief synopsis of the proposed solution.

Note: A one-page PDF or JPEG file can be uploaded to support your synopsis. (Max. 5MB)

Project Synopsis

Multi-sensor fusion — combining radar, imagery, signals and acoustic data into one picture — is how modern forces make sense of contested environments. But every data element carries classification rules, release authorities and coalition caveats, and today those are enforced by manual review and checklists that cannot keep pace with AI-driven fusion. Canada lacks a sovereign, Canadian-controlled engine to enforce them automatically. Planetar's Fusion Compliance Engine (FCE) closes that gap. It sits at the single message chokepoint every fused observation crosses and, in real time, evaluates machine-readable policy against each data element — deciding whether to permit, restrict, downgrade or segregate it before a cross-domain merge occurs — while writing an immutable, exportable audit-and-lineage record. It enforces compliance during fusion without adding latency that would degrade a tactical decision. At project start the supporting platform is working open-source code: a provenance-tracked real-time message bus with an append-only audit log, a typed envelope that already carries source and timing provenance, a machine-readable policy layer (W3C ODRL), a per-edge lineage graph, an analyst console, and four per-modality detectors. The new research for Component 1a is the policy-evaluation-and-enforcement engine itself, the classification and domain provenance fields, the enforcement-action audit record, the exportable lineage format, and an FPGA-prototyped hardware enforcement datapath for edge deployment. End-state: an evaluator-operable live demonstration at planetar.ca that fuses two modalities (AIS and synthetic-aperture radar) under synthetic Protected-B markings, enforces policy per element, and exports an accreditation-grade lineage trail. A two-person Zax Analytics team; public data only; no government-furnished property, no field deployments, no real classified content.

B. Keywords

Provide 3 to 5 keywords related to your project to help us index the solutions and assign the appropriate technical and scientific evaluators to the proposal.

Keyword
multi-sensor data fusion
compliance-by-design
automated policy enforcement
data provenance and audit lineage
machine-readable classification policy (ODRL)

C. Project Overview

Provide a project overview, which is a synopsis of project's S/T Merit, Novelty and Innovation, Impact, Feasibility and Approach.

The Fusion Compliance Engine (FCE) is a modular, AI-enabled component that enforces classification and release-authority policy in real time during multi-sensor fusion. It sits at the single message chokepoint between sensor ingestion and the fusion pipeline, evaluates machine-readable policy against each data element, and decides — permit, restrict, downgrade or segregate — before a cross-domain merge occurs, while emitting an immutable, exportable audit-and-lineage record. The supporting real-time, provenance-tracked platform is already built and open-source. Scientific and Technical Merit (PRC-1): the science is per-element policy enforcement inside the fusion path, at the chokepoint, fast enough to add no tactical latency. It rests on recognised standards (W3C ODRL policy; NATO STANAG 4774/4778 classification markings; W3C PROV lineage), an attribute-based decision over each element's classification, domain, source, modality and releasability, and a built, measured nanosecond-scale message bus with an immutable write-ahead log. Novelty and Innovation (PRC-2): compliance for fusion is enforced today by manual review or by perimeter cross-domain guards that screen at a network boundary. The FCE instead renders per-element dispositions in-path, during fusion — inherently always-invoked and non-bypassable because every element crosses one bus — and repurposes a civil machine-readable rights standard (ODRL) as sovereign defence classification policy. Accreditation-grade lineage is a structural property of the substrate, not a bolt-on report. Impact (PRC-3): the FCE closes a stated sovereign capability gap, removing the manual-review bottleneck so cross-domain fusion is both faster and audit-safe. It is a reusable building block for every future fusion system — from Arctic surveillance to coalition interoperability — built on the applicant's own open-source code and net-new, Canadian-owned IP, aligned to NORAD modernization, the CAF Digital Campaign Plan, and the DND/CAF AI Strategy. Feasibility and Approach (PRC-4): the work runs over two milestone stages on public data, adding enforcement to a chokepoint the applicant already owns. Milestone 1 builds the policy model, the additive classification and domain fields, the demonstration data setup, and the architecture of a hardware enforcement datapath. Milestone 2 builds the enforcement engine, the write-ahead-log enforcement record, the exportable lineage, an evaluator-operable live planetar.ca demonstration on two modalities under synthetic Protected-B markings, the enforcement-overhead benchmark, and an FPGA prototype of the datapath (a RISC-V core with custom policy-check instructions) for edge and server profiles. A two-person Zax Analytics team (founder + IC-design engineer); no government-furnished property, no field deployments, no real classified content.

MC-1 - Technology Readiness Level (TRL) of the Proposed Solution

Select and describe the TRL of the proposed solution before starting the project and the projected end-state under the contract.

Note: [Information on Technology Readiness Levels \(TRLs\)](#).

TRL of the solution before starting the project: **TRL-2: Technology concept and/or application formulated**

Describe the research and development activities that have taken place to bring the proposed solution to the stated TRL.

Current TRL: 2. The solution is a modular Fusion Compliance Engine (FCE) that evaluates machine-readable classification and release-authority policy at the single message chokepoint between sensor ingestion and fusion, deciding per data element whether to permit, restrict, downgrade, or segregate, and emitting an immutable, exportable audit-and-lineage record. The concept is fully formulated with strong component-level evidence from the research and development below, but the enforcement engine that binds policy to dispositions is not yet built or demonstrated; building and demonstrating it is the work the project funds, advancing the solution to TRL 3. The fusion chokepoint is already built and open-source: planetar-broker, a dependency-light single-file C message bus (TCP, UDP, and shared-memory transports), is the single point every fused observation crosses. Its append-only write-ahead log is CRC32-protected and bit-exact replayable — the audit-and-lineage spine the FCE writes to. A one-million-message benchmark of planetar-broker's own shared-memory ring recorded p50 95–100 ns and p99 1.0–1.6 μ s across four runs on an untuned workstation; measuring the FCE's enforcement overhead against this baseline is part of the project, and that headroom shows a policy check adds no material tactical latency. Native provenance is built: zmesg, a zero-copy typed envelope, already carries a source identifier, nanosecond timestamps, and correlation and causation ids on every message — most of the provenance record an Essential Outcome requires. Classification marking and domain of origin are small additive fields delivered in the project. Machine-readable policy is built: planetar-market ships a working W3C ODRL policy layer — a standards-aligned rights language, with tests — that expresses machine-readable terms but, by its own design, does not yet enforce them. Adopting ODRL as the FCE's classification and release-authority language, and building the engine that acts on it, is the central advance of the project. Per-edge lineage and the analyst surface are built: planetar-ontology, a zero-dependency, test-covered service, tracks provenance per graph edge from ingestion through fused output, and planetar-ui renders each result's causal evidence chain. This entity-resolution-with-provenance work is a substantial development beyond the integrated-entity-view architecture of US Patent 10,936,582, on which the applicant is a named inventor (Salesforce-assigned), not the patent holder. The applicant's peer-reviewed semi-supervised machine learning on scarce-label sensor streams (ORCA-SLANG 2021; Sattar et al. 2011, on Ocean Networks Canada data) evidences the capability to deliver. Together this work formulates and de-risks the concept — chokepoint, provenance, policy language and lineage all exist — so the six months is spent on the enforcement engine itself, advancing the solution from TRL 2 to TRL 3.

Select the projected end-state TRL of the proposed solution under the contract.: **TRL-3: Analytical and experimental critical function and/or characteristic proof of concept**

Provide a high-level explanation with relevant examples to justify your anticipated TRL at the end-state by applying the TRL definitions provided above.

At project start the solution — a modular Fusion Compliance Engine — is at TRL 2: the concept is formulated with strong component-level evidence (a built, provenance-tracked fusion chokepoint with an immutable audit log, a built machine-readable policy layer, per-edge lineage, and an analyst console), but the enforcement engine that binds policy to real-time dispositions is not yet built or demonstrated. The anticipated end-state is TRL 3 — analytical and experimental critical-function proof of concept — reached when the engine is built and demonstrated end-to-end. The demonstration is twofold. First, a controlled, reproducible scenario in a Maritime Domain Awareness setting: two sensor modalities — open AIS (unclassified) and Sentinel-1 synthetic-aperture-radar detections marked at synthetic Protected B — are fused; as each element crosses the chokepoint the FCE evaluates the machine-readable policy and applies a disposition (permit, restrict, downgrade, or segregate), blocking an unauthorized cross-domain merge, writing an enforcement-action record to the immutable log, and producing an exportable ingestion-to-output lineage trail, all replayable bit-for-bit from the write-ahead log. Second, an evaluator-operable live system at planetar.ca that evaluators drive themselves during evaluation — changing a policy and watching enforcement hot-reload, and clicking any disposition back through its governing rule and provenance. A running system an evaluator can operate is the defining characteristic of a demonstrated proof of concept. The work proceeds in two milestone stages. Milestone 1 designs the ODRL-to-classification policy model, adds the classification and domain-of-origin fields to the envelope, designs the synchronous policy-evaluation hook, sets up the two-modality demonstration data under synthetic Protected-B markings, and specifies the hardware enforcement-datapath architecture with simulation. Milestone 2 builds the enforcement engine at the chokepoint, adds the enforcement-action record type to the write-ahead log, delivers the exportable lineage format and the evaluator-operable planetar.ca demonstration, measures the FCE's enforcement overhead against the no-enforcement baseline, and demonstrates an FPGA prototype of the enforcement datapath (a RISC-V core with custom policy-check instructions) on live bus traffic. Every disposition traces to the rule that produced it and to the element's provenance, and the full chain can be independently re-run from the open-source repositories — analytical and experimental evidence, not assertion. This is a clean one-level advance from TRL 2 to TRL 3, squarely inside the Component 1a bracket (TRL 1–3). TRL 4 — component validation in a laboratory environment — is the natural Component 1b objective this proof of concept sets up.

MC-2 - Alignment of Proposed Solution to S&T Challenge

Describe the proposed solution and how it addresses and aligns to the S&T Challenge identified above by including a description of the scientific and technological basis upon which the solution is proposed, and clearly justify how the solution meets the Essential Outcomes.

MC-2 - Alignment of Proposed Solution to S&T Challenge

Solution. The Fusion Compliance Engine (FCE) is a modular, AI-enabled compliance layer that sits between sensor ingestion and the fusion analytics pipeline and acts as a policy-aware gatekeeper: it tags, filters, and routes each data element by machine-readable classification and operational-release policy, in real time, during multi-sensor fusion — exactly the automated compliance layer CH14 requires in place of manual review and procedural checklists. It is built on a provenance-tracked message bus the applicant has already developed, and demonstrated in a Maritime Domain Awareness setting (a CH14 example) fusing open AIS and synthetic-Protected-B SAR detections, so cross-domain merges carry a genuine classification decision. Essential Outcome compliance: - (EO1) A modular AI-enabled component (knowledge-based AI: automated reasoning over machine-readable policy) enforces classification and policy constraints during multi-sensor (at least two) fusion — enforcement is a synchronous hook at the bus chokepoint every fused element already crosses. - (EO2) Enforcement is driven by machine-readable policy (W3C ODRL [P1], already built as an expression layer) applied across two sensor modalities (AIS, SAR), at least the Network-security domain (each element's originating security domain travels in the envelope's domain-of-origin field), and at least Protected B (the Government of Canada categorization baseline [S6]); classification markings are structured to NATO STANAG 4774 confidentiality-label syntax [S1]. - (EO3) Checks and enforcement actions run programmatically during ingestion and fusion, without human approval for predefined policy conditions. - (EO4) Provenance records are generated and retained for all data in and out of the pipeline — source-sensor id, classification marking, timestamp, and domain of origin (zmesg envelope + the 1a's added classification/domain fields), each marking bound to its element per NATO STANAG 4778 [S2]. - (EO5) Audit logs record the policy rules applied, the enforcement action taken (permit / restrict / downgrade / segregate), and the resulting disposition, written to the CRC32 append-only WAL. - (EO6) Data lineage from original ingestion through fused output is reconstructable (WAL replay + per-edge graph provenance) and exportable for compliance review, forensic analysis, or accreditation. Fit to the challenge intent. CH14 seeks a sovereign, Canadian-developed and Canadian-controlled compliance engine for multi-domain fusion — a reusable building block "from Arctic surveillance to coalition interoperability hubs," supporting the CAF Digital Campaign Plan, the DND/CAF AI Strategy, NORAD modernization, and the Cyber Forces mandate. The FCE is Canadian-owned foreground IP built on the applicant's open-source spine; because it enforces at a chokepoint already benchmarked at nanosecond scale, it adds compliance without degrading tactical decision speed.

PRC-1 - Scientific and/or Technical (S/T) Merit of Proposed Solution

Articulate how your proposed solution meets the Scientific and/or Technical Merit criterion.

PRC-1 - Scientific and/or Technical (S/T) Merit of Proposed Solution

(a) Sound S&T evidence. The 1a's science — real-time, per-element policy enforcement inside a multi-sensor fusion pipeline — rests on recognised standards, working implementations, and the applicant's peer-reviewed record. The component is itself AI-enabled, per the CH14 Q&A's explicit acceptance of knowledge-based AI: the engine is automated reasoning over a machine-readable policy knowledge base, and it governs the learned per-modality detections whose outputs it fuses — with every disposition auditable by construction. Machine-readable policy: W3C ODRL [P1], a standards-track rights language used across IDSA, Gaia-X, and JPEG Trust, supplies the policy vocabulary; the applicant has already built an ODRL expression layer (planetary-market, test-covered). Attribute-based decisioning: mapping a policy to a decision over an element's {classification, domain, source, modality, releasability} is an attribute-based formulation [P4] applied in-path rather than at an access gate. Classification + provenance standards: the design targets NATO STANAG 4774 confidentiality-label syntax [S1] with STANAG 4778 binding [S2], and models exportable lineage on W3C PROV [S5] — interoperable foundations, not bespoke markings. Engineering evidence: the enforcement seat is a built, provenance-tracked single-file C bus whose shared-memory chokepoint is benchmarked at p50 95–100 ns / p99 1.0–1.6 μ s (four 1M-message runs), with a CRC32 append-only WAL for immutable audit. Pedigree: the applicant's peer-reviewed ML on continuous, scarce-label sensor streams [A1, A2] and productionised entity-resolution-with-provenance experience evidence the capability to deliver a measurable system. (b) State of the art. The FCE advances current practice on three axes. Enforcement placement: incumbent policy languages decide access at a decision point (XACML/ABAC [P3, P4]) and cross-domain guards screen at a network boundary (NCDSMO "Raise the Bar" [S3]); the FCE instead renders per-element dispositions — permit/restrict/downgrade/segregate — in-path during fusion, inherently always-invoked and non-bypassable because every element crosses one bus. Expression vs enforcement: ODRL [P1] expresses rights but, by design, does not enforce them; the 1a's evaluation-and-enforcement engine binding an ODRL policy to a real-time disposition is the advance. Lineage as a property, not a report: because each decision is journalled to a replayable CRC32 WAL and every fused edge carries provenance, accreditation-grade lineage is structural — beyond the post-hoc audit that current manual-review compliance produces. Falsifiable 1a outcomes: (i) FCE-on enforcement overhead vs. the no-FCE baseline; (ii) policy-decision correctness on a labelled synthetic Protected-B set; (iii) lineage-export fidelity — reconstructed ingestion-to-output chain vs. ground truth.

PRC-2 - Novel & Innovative Solution

Articulate how your proposed solution meets the Novel and Innovative criterion.

PRC-2 - Novel & Innovative Solution

(a) New knowledge / technology. (1) Enforcement embedded in the fusion path — the core contribution. Compliance for multi-domain fusion is enforced today by manual review and checklists, or by cross-domain guards [S3] that screen at a network boundary, beside or after fusion. The FCE instead places a synchronous policy gate at the single message chokepoint every fused element crosses, deciding per element to permit, restrict, downgrade, or segregate before a cross-domain merge occurs. Per-element, in-line enforcement during multi-sensor fusion — not perimeter screening or post-hoc audit — is, to the applicant's knowledge, absent from the open literature: novel, separately patentable Canadian foreground IP. (2) Machine-readable rights repurposed as defence classification policy. The FCE adopts W3C ODRL [P1] — a standards-track rights language used across IDSA, Gaia-X, and JPEG Trust — as the policy language for classification guides, release authorities, and coalition caveats. The applicant has built an ODRL expression layer (planetary-market) that by design expresses but does not enforce; the novelty is the evaluation-and-enforcement engine binding an ODRL policy to a real-time disposition over an element's {classification, domain, source, modality, releasability}, hot-reloadable with no restart. This brings a civil-standards substrate to sovereign defence fusion. (3) Accreditation-grade lineage as a property of the substrate. Because every decision is written to an append-only, CRC32-protected, replayable log and every fused edge carries provenance, exportable ingestion-to-output lineage is a structural property, not a bolt-on report — forensic review and accreditation by construction. (b) Enhanced capability vs SOTA. Speed: manual compliance cannot keep pace with AI-fusion volume; the FCE enforces at a benchmarked chokepoint (p50 95–100 ns / p99 1.0–1.6 μ s), adding negligible tactical latency — the CH14 constraint. Placement: in-path per-element dispositions (downgrade/segregate, not just allow/deny) versus boundary guards' coarse pass/block. Trust: each disposition renders its governing rule and provenance for inspection and controlled override, beyond opaque after-the-fact logging. Pedigree: the applicant's peer-reviewed reliable-ML work on noisy, scarce-label sensor streams [A1, A2] and named-inventor provenance/entity-resolution IP (US 10,936,582 — Salesforce-assigned background, not owned) evidence capability to deliver. (c) Future potential. The "enforce policy at the fusion chokepoint" pattern generalizes to any multi-domain setting CH14 names — Joint ISR (SIGINT + EO/IR + radar), Arctic surveillance, tactical edge, coalition interoperability — without redesign, since the engine is agnostic to which modalities cross the bus. As sovereign Canadian foreground IP, it is a reusable building block for every future fusion system and a basis for Component 1b/2 hardening.

PRC-3 - Impact of Proposed Solution

Articulate how your proposed solution meets the Impact criterion.

PRC-3 - Impact of Proposed Solution

(a) Addresses a stated capability gap. CH14 states the gap plainly: Canada "lacks a sovereign, Canadian-developed and Canadian-controlled compliance engine for multi-domain fusion," and classification enforcement during fusion is done today by manual review and procedural checklists that cannot keep pace with AI-fusion volume and velocity. The FCE removes that bottleneck — machine-readable policy is enforced automatically, per element, at the chokepoint between ingestion and fusion — delivering CH14's two operational benefits at once: it reduces risk (no un-audited or non-compliant cross-domain merge passes silently; every disposition is journalled immutably) and increases tempo (enforcement runs at fusion speed, without the human-approval step that gates manual review). The capability is built on the applicant's own AGPL open-source spine and net-new, Canadian-owned FCE foreground IP — a genuinely sovereign capability, replicable on commodity Linux, aligned to NORAD modernization, the CAF Digital Campaign Plan, the DND/CAF AI Strategy, and the Cyber Forces mandate. (b) Enhances S&T capability. Automated, in-path, per-element compliance enforcement during multi-sensor fusion is a capability not present in current Canadian fusion systems. Accreditation by construction: exportable, replayable ingestion-to-output lineage (modelled on W3C PROV [S5]) supports compliance review, forensic analysis, and accreditation without a bolt-on reporting step. Coalition interoperability: classification markings structured to NATO STANAG 4774/4778 [S1, S2] let releasability and coalition caveats be enforced and shared by machine, not memo. Operator trust: every disposition renders its governing rule and the element's provenance for inspection, with controlled, immutably-logged override — addressing a known barrier to operational AI adoption. (c) Matures the field. The FCE is, in CH14's words, "a reusable building block for every future fusion system." Generalizable pattern: "enforce policy at the fusion chokepoint" transfers without redesign to Joint ISR (SIGINT + EO/IR + radar), Arctic surveillance, tactical-edge dismounted, and coalition interoperability hubs — the engine is agnostic to which modalities cross the bus. Lower barrier to creditable AI: by making compliance a structural property rather than an afterthought, the FCE shortens the path to authority-to-operate for fielded AI fusion. Durable, sovereign output: applicant-controlled open-source code and new Canadian-owned IP outlast the contract window as reference implementations and a publishable contribution to CAF multi-domain compliance.

PRC-4 - Feasibility and Approach of Proposed Solution

Articulate how your proposed solution meets the Feasibility and Approach criterion.

PRC-4 - Feasibility and Approach of Proposed Solution

Starting position de-risks the build. The FCE is not built beside a fusion system; it is enforcement added at a chokepoint the applicant already owns and operates. Built, open-source, and broker-integrated today: planetar-broker (the C message bus and its CRC32 append-only WAL), zmesg (the typed provenance envelope), planetar-market (the W3C ODRL policy-expression layer, test-covered), planetar-ontology (per-edge lineage), planetar-ui (the analyst shell), and four per-modality ingest/detection services — planetar-ais (live AIS), planetar-sat (Sentinel-1 SAR with CFAR detection on a real Sentinel-1C scene), planetar-eo, and planetar-acoustic. The substrate already carries provenance and journals every message immutably; the 1a adds the enforcement engine, not the pipeline. Scoped, bounded 1a build. Six bounded increments: (1) add classification-marking and domain-of-origin fields to the zmesg envelope; (2) a synchronous policy-evaluation hook in the broker publish/subscribe path; (3) the ODRL-to-classification policy model with permit/restrict/downgrade/segregate dispositions and hot-reload; (4) an enforcement-action record type in the WAL; (5) an exportable ingestion-to-output lineage format; (6) a hardware enforcement datapath — a RISC-V core with custom instructions for envelope parse, label match, and disposition — simulated and FPGA-prototyped for server and sensor-adjacent (camera/UAS) profiles. Latency feasibility — the FCE's key risk — is already evidenced. The challenge forbids latency that degrades tactical decisions. The bus's shared-memory path is benchmarked at p50 95–100 ns / p99 1.0–1.6 μ s over four 1M-message runs, leaving ample headroom to evaluate a policy and write a disposition. The 1a will measure FCE-on enforcement overhead against this baseline and report it conservatively (bracketing measured runs, no post-tuning figures). Data feasibility. DND/CAF supply no data; the applicant already generates the needed feeds (live AIS, Sentinel-1 SAR via Copernicus) and will apply synthetic classification markings (open AIS unclassified, SAR detections marked Protected B) so the demonstrator exercises a genuine cross-domain-merge decision with no real classified material. Team and plan. A two-person team delivers the 6-month 1a: the founder (PhD, CS/ML), who built the entire spine, and IC-design engineer Colin Broughton — named inventor on US patents including a multithread embedded processor (US 7,320,065) — owning increment (6). Milestone 1: policy model, envelope fields, demo/data setup, datapath architecture + simulation. Milestone 2: enforcement engine, WAL audit record, exportable lineage, evaluator-operable planetar.ca demo, overhead benchmark, and the FPGA datapath prototype on live bus traffic; Milestone 1 is 39.8% of the total, under the 70% CFP cap. Software risk is additive integration onto a working substrate; the hardware package is scoped to FPGA proof, not silicon.

PRC-5 - Gender-based Analysis Plus (GBA Plus) of Proposed Solution

Articulate how your proposed solution meets the Gender-based Analysis Plus criterion.

Note: [Additional information on Gender-based Analysis Plus \(GBA Plus\).](#)

PRC-5 - Gender-based Analysis Plus (GBA Plus) of Proposed Solution

GBA Plus is applied to the FCE's technical solution along three axes, with one cross-cutting principle: the engine enforces predefined policy automatically, and the human analyst stays accountable — through review, explanation, and logged override — guarding against automation bias that can fall unevenly on under-represented groups. (i) Bias in automated enforcement — the FCE-specific axis. An engine that automatically permits, restricts, downgrades, or segregates each data element exercises consequential authority; if its policy or the sensor coverage behind it is uneven, that authority falls unevenly. Classification and release rules — and the modality coverage informing them — are denser for commercial shipping lanes and Northern-hemisphere SAR than for Indigenous coastal fishing waters or low-traffic Arctic regions, so an FCE could systematically over-segregate (excluding under-represented sources from fusion) or mishandle data tied to specific communities. The 1a treats this as measurable: a disposition-audit protocol stratifies enforcement decisions by source, modality, and region to surface uneven handling — a concrete artefact, not a generality. Because every disposition is written human-readably to the immutable WAL with its governing rule and provenance, each decision is reviewable and contestable, and an authorised operator can override it with the override itself logged accountably. (ii) Operator-side accessibility and multilingual usability — an M2 deliverable. CAF operators span genders, ages, abilities, neurodivergence, and linguistic backgrounds. The shell (planetar-ui, React 19) that surfaces each compliance decision and its causal evidence chain is built against a documented accessibility checklist: keyboard navigation, ARIA roles on the decision/provenance panels, screen-reader compatibility, and a colour-vision-deficient-safe palette (disposition double-encoded with shape/label, not colour alone). Bilingual-ready English/French via i18n. Plain-language, explainable decisions plus controlled override lower the expertise barrier, widening who can operate the system beyond specialist compliance officers. The accessibility checklist is itself a 1a deliverable. (iii) Deployment-side equity. The engine's SWaP profile — commodity-Linux, laptop-class, no kernel bypass or FPGA required (the single-file C broker) — means it deploys at the same fidelity in low-bandwidth or remote settings (Indigenous coastal communities, small-port operations, Arctic coast-guard auxiliary stations) as in central operations centres. Equity is a structural property of the architecture, not a deployment afterthought. Implementation in the 1a: the disposition bias-audit protocol (M2 evaluation) and the operator-accessibility + bilingual-readiness checklist (M2 demo) are explicit work-plan deliverables — GBA Plus applied to the technical solution, not deferred to a follow-on phase.

PRC-6 - Alignment of Desired Outcomes

Articulate how your proposed solution meets the Alignment of Desired Outcomes criterion.

Note: If no Desired Outcomes are identified in the Challenge statement, any information provided by the Offeror in this section will not be evaluated, and Offerors will receive 15 points.

PRC-6 - Alignment of Desired Outcomes

(DO1) Real-time enforcement at tactical-decision speed. The FCE enforces at the message chokepoint every fused element already crosses, benchmarked at p50 95–100 ns / p99 1.0–1.6 μ s on its shared-memory path over four 1M-message runs — orders of magnitude under any tactical-decision threshold. Because the policy check is synchronous and in-path, enforcement is real-time by construction across the demo's two modalities (AIS, SAR) and Protected B markings. The 1a measures FCE-on overhead against this baseline and reports it conservatively (bracketing measured runs, no post-tuning figures), so the "no added tactical latency" outcome is evidenced, not asserted. (DO2) Adaptable policy without restart. FCE policy is W3C ODRL [P1] — data, not code — so classification guides, release authorities, and coalition caveats are reconfigured by swapping a policy set, not recompiling. The 1a demonstrates a live policy hot-reload mid-session (e.g. tightening a releasability caveat) with enforcement continuing uninterrupted: the adaptable-framework outcome shown on a running system, not described. (DO3) SWaP / edge deployment. The enforcement substrate is a dependency-light, single-file C broker (~1.5k LOC, half again as much in doc comments) that runs on commodity hardware — directly the CH14 Tactical-Edge / Dismounted example (a compliance layer on a ruggedized laptop or edge server). The 1a states the measured compute and memory envelope of FCE-on operation, showing enforcement is maintained within an edge SWaP budget rather than requiring datacentre resources. It also delivers an FPGA-prototyped hardware enforcement datapath — a RISC-V core with custom instructions for envelope parse, label match, and disposition — evidencing a path to sensor-adjacent enforcement on cameras and uncrewed platforms, and to server-side offload, within the same SWaP envelope. (DO4) Explainability + controlled override. Every disposition is rendered for the operator: planetar-ui drills from a compliance decision back through its causation chain to the governing rule and the element's provenance, so an analyst sees why data was permitted, restricted, downgraded, or segregated. Override is a first-class, accountable action — an authorized operator can override a disposition, and the override is itself written to the CRC32 append-only WAL as an immutable, attributable enforcement record, satisfying the accountability-safeguard requirement.

PRC-7 - Alignment of Proposed Solution Costs

In the following sections, you will be asked to provide your detailed work plan and financial proposal. The breakdown of the Financial Proposal must align with the work proposed, and specifically:

- a. Is realistic for the technical approach proposed;
- b. Is appropriate with, and is reasonably relative to:
 - i. each labour category proposed per milestone;
 - ii. the number of labour hours proposed per milestone;
 - iii. the types and categories of materials, equipment, and other direct costs;
 - iv. each task for which travel and costs incurred is proposed, the travel costs incurred are explained and reasonable.

Solution Progression to Follow-on Components

Zax Analytics intends to progress this solution into Component 1b (Develop, TRL 4-5) and beyond. Component 1a delivers a demonstrated proof-of-concept at TRL 3: a modular Fusion Compliance Engine (FCE) that, at the single message chokepoint

between sensor ingestion and fusion, evaluates machine-readable classification and release-authority policy against each data element in real time — permitting, restricting, downgrading or segregating it — while emitting an immutable, exportable audit-and-lineage record. It is demonstrated across two sensor modalities under synthetic Protected-B markings and is operable live at planetar.ca, where a Technical Authority can drive it during evaluation, change a policy and watch enforcement hot-reload; the 1a also proves the enforcement datapath in hardware, as an FPGA-prototyped RISC-V core with custom policy-check instructions. S&T work under Component 1b advances the capability from proof-of-concept to operationally-relevant maturity along four lines. 1) Productionize and harden the engine. We move from synthetic markings to operationally representative classification guides, release authorities and coalition caveats; harden the policy-evaluation path for throughput, graceful degradation when sensors drop, and resistance to adversarial inputs and policy-spoofing. The controlled, audited operator override captured in 1a becomes an online operator-correction loop, so analyst adjudications continuously refine the policy model while every change remains logged and attributable. 2) Operational validation with a CAF partner. We validate the FCE against operator-defined classification policy on representative multi-domain data in a controlled environment (TRL 4), then in an operationally relevant environment (TRL 5). Every disposition remains traceable to the rule that produced it and to the element's provenance, so results are accreditable rather than opaque, and the enforcement overhead is measured against the no-enforcement baseline to confirm no tactical-decision latency is added. 3) Extend across modalities and domains. The core idea — enforcing policy in-path at the one chokepoint every fused element crosses — generalizes without redesign, because the engine is agnostic to which modalities cross the bus. From the maritime beachhead it extends to Joint ISR fusion (signals intelligence, electro-optical/infrared imagery and radar tracks), to Arctic surveillance, and to coalition interoperability hubs, governing additional security domains and classification levels beyond the Network-security and Protected-B demonstrated in 1a. 4) Edge deployment and accreditation. We adapt the dependency-light, single-file engine for size-, weight- and power-constrained tactical-edge hardware (the Tactical-Edge / Dismounted use case) and degraded-connectivity operation; align the confidentiality markings and their binding to NATO STANAG 4774 and 4778 for coalition interoperability; add multi-level handling so the system enforces across security levels with full lineage; and begin a formal accreditation / Authority-to-Operate readiness package built on the exportable audit-and-lineage trail. In parallel, the 1a's FPGA-proven enforcement datapath is hardened toward a test-chip tape-out via subsidized Canadian fabrication-access programs (e.g. CMC Microsystems multi-project-wafer runs), targeting a server offload card and a sensor-adjacent enforcement module for camera and uncrewed platforms — sovereign silicon for the same chokepoint. End-state of Component 1b: a validated Fusion Compliance Engine at TRL 5 — component and system validated in a relevant environment — enforcing real machine-readable classification and release-authority policy across at least two security domains, with an accreditation-readiness package and a clear path to TRL 6+ operational deployment. Capabilities and benefits for Canada. The FCE closes a capability gap the challenge states plainly: Canada lacks a sovereign, Canadian-developed and Canadian-controlled compliance engine for multi-domain fusion. It removes the manual-review bottleneck so cross-domain fusion is both faster and audit-safe, and it is a reusable building block for every future fusion system — from Arctic surveillance to coalition interoperability hubs — supporting NORAD modernization, the CAF Digital Campaign Plan, the DND/CAF AI Strategy and the Cyber Forces mandate. Because it is grounded in the applicant's own open-source code and net-new, Canadian-owned foreground IP, and is replicable on commodity hardware, it is a

sovereign Canadian capability that interoperates with allies. Advancement to market and commercialization. The capability is dual-use and broadly reusable. Beyond the CAF, the same compliance-and-provenance engine serves civil multi-domain awareness for the Canadian Coast Guard, Fisheries and Oceans Canada, Transport Canada and port authorities, and allied defence and security customers governing coalition data exchange. Zax Analytics will mature the 1a and 1b deliverables into a deployable product under a permissive-substrate, dual-licensed-platform model that preserves the commercial

upside. The progression path is Component 1a (Conceive, TRL 3), to Component 1b (Develop, TRL 4-5), to operational deployment and procurement (TRL 6+), positioning Canada with a sovereign, exportable compliance engine for every future fusion system.

Work Plan and Deliverables

This section will be used to assess the criteria and supporting information of PRC-4.

Each row of the following tables must contain specific activities and deliverables (work). Work must be distributed into two (2) milestone stages. Work for each milestone stage must render the project to a point of progress sufficient to enable the Technical Authority (TA) to assess the potential of the Project Description through completion of each Milestone thereafter.

In the tables that follow, Offerors must specify the work activities to be performed. Decompose the work activities to a level that exposes all risk factors, that allows accurate estimation of the requirements, and the schedule duration for each task activity (**maximum 800 characters per table entry**).

Milestone tables

Funds are limited to a maximum allowance of **CAN\$250,000** (excluding applicable taxes) per contract for a performance period of up to six (6) months.

The cost proposal for Milestone 1 must be no greater than **70%** of the total cost for the proposed solution. A table must be filled in for each Milestone.

Financial Proposal

This section will be used to assess the criteria and supporting information for SC-1 and PRC-7.

In the tables that follow, Offerors must specify the various costs proposed for each milestone (excluding applicable taxes).

The two (2) milestone tables must be completed. Cost(s) indicated in the "Other Costs" table must fall within the definition of Eligible Costs in Section 3.7 of the Solicitation document.

The total cost proposal must not exceed the individual contract maximum of **CAN\$250,000** (excluding applicable taxes).

The Total Firm Milestone 1 Price must be no greater than **70%** of the total cost proposal (Milestones 1 & 2 combined). The Financial Proposal must align with the work proposed.

Note: All costs identified in the proposal must be in Canadian dollars.

Milestone 1

Milestone 1 requires the following:

Progress reporting (Milestone 1) to be completed and delivered in accordance with the Statement of Work (SOW).

Total Performance Period (in weeks) 13 (weeks)

Work Activities & Risks and Mitigation

Activity	Description of Deliverable	Estimated Level of Effort (Weeks)	Risk(s): Description, probability and impact	Risk mitigation strategy(ies)
Design the FCE policy model: map classification guides, release authorities and coalition caveats onto W3C ODRL constructs, defining the disposition vocabulary (permit, restrict, downgrade, segregate) as a decision over each element's classification, domain, source, modality and releasability. Policy sets are versioned data, loadable at runtime (hot-reload, no restart). Author the synthetic Protected-B policy set the demonstration enforces.	A documented ODRL-based classification/release policy model with the disposition vocabulary, plus the versioned synthetic Protected-B policy set used by the demonstrator.	3.00	ODRL was designed to express digital rights, and some defence classification semantics (caveats, releasability chains) may not map cleanly onto its core constructs (medium probability). Impact: policy model rework within this activity.	The applicant already ships a working ODRL expression layer (planetary-market, test-covered); extend it with a defence-classification ODRL profile rather than bending core constructs, and anchor the marking vocabulary to NATO STANAG 4774 label syntax so the semantics follow an established standard.
Add the classification-marking and domain-of-origin fields to the zmsg typed envelope — structured to NATO STANAG	A versioned envelope carrying		Envelope changes touch every service codec, so an	The fields are additive and the envelope is versioned; every codec is in

4774 confidentiality-label syntax, bound to the element per STANAG 4778 — and update the per-service codecs so every message on the bus carries its marking end-to-end, under regression tests.	classification-marking and domain-of-origin fields on every message, with all in-repo service codecs updated and regression gates passing.	2.00	inconsistency could ripple across the pipeline (low-medium probability). Impact: integration rework contained to the codecs.	applicant-controlled open-source repositories, and the change is locked behind the existing regression suites before any downstream work depends on it.
Design the synchronous policy-evaluation hook at the broker publish path: evaluation order, fail-safe semantics (default-deny on policy fault), the four disposition actions, and the enforcement-action record type written to the write-ahead log. Define the FCE-on overhead measurement protocol against the measured shared-memory baseline (p50 95-100 ns / p99 1.0-1.6 microseconds, four 1M-message runs, 2026-07-13).	The enforcement-hook design specification: evaluation semantics, fail-safe behaviour, disposition actions, WAL enforcement-action record schema, and the overhead measurement protocol for Milestone 2.	2.00	A synchronous in-path check could add measurable latency to the fusion path (medium probability). Impact: erodes the no-added-tactical-latency outcome if unmanaged.	Design against the measured nanosecond-scale baseline: policies compile ahead of time to a fast lookup at publish time, with the rare complex case escalated rather than evaluated inline. Overhead is then measured in Milestone 2 and reported conservatively — bracketed measured runs, no post-tuning figures.
Set up the two-modality demonstration				

data: live open AIS and Sentinel-1 SAR detections, with synthetic classification markings applied (AIS unclassified; SAR detections marked Protected B) so the demonstrator exercises a genuine cross-domain-merge decision without touching real classified material. Package the scenario for bit-exact replay from the write-ahead log.	A replayable AIS + SAR demonstration dataset under synthetic Protected-B markings — the scenario the Milestone 2 demonstrator and evaluation run against.	2.00	Sentinel-1 scene availability or processing for the chosen operating area could lag (low probability). Impact: demo data setup slips within the milestone.	The SAR pipeline (planetar-sat: fetch, CFAR detect) is already built and validated on a real Sentinel-1C scene; archived scenes are an equivalent fallback, since the compliance decision is driven by the synthetic markings, not scene freshness.
Specify the hardware enforcement-datapath architecture: a RISC-V core extended with custom instructions for envelope parse, classification-label match and disposition, targeting two profiles — server offload at the chokepoint and sensor-adjacent enforcement on cameras and uncrewed platforms. Validate the design in cycle-level	The datapath architecture specification (instruction set, pipeline, both deployment profiles), cycle-level simulation results against recorded bus traffic, and the	4.00	A fixed custom-instruction set may not cover every policy-match case in hardware (medium probability). Impact: some policy classes would fall back to software evaluation, reducing the offload benefit	Hybrid by design: common-case dispositions execute in hardware; rare or complex policies escalate to the software FCE, which remains the complete reference implementation. The hardware package is scoped to architecture-and-FPGA

simulation against recorded bus traffic and define the Milestone 2 FPGA prototyping plan. Owned by the IC-design engineer.	Milestone 2 FPGA prototype plan.		for those cases only.	proof — the software engine never depends on it.
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Milestone 1 - Labour			
Category	Labour (h)	Rate (\$/h)	Total (\$)
Principal Investigator (PhD, Computer Science / Machine Learning)	400	140.00	56000.00
IC-Design Engineer (hardware enforcement datapath)	100	140.00	14000.00
Total Labour Cost (\$)			70000.00

Milestone 1 - Other Costs		
Other cost	Description	Cost (\$)
Cloud compute	Cloud compute for Sentinel-1 scene fetching and processing, policy-evaluation test runs, and hosting the live planetar.ca system. No model training.	1500.00
Software / tools	Development and operations tooling: continuous-integration runner minutes, observability/monitoring, and ancillary paid developer services. RTL work uses open-source toolchains.	500.00
Total Labour Cost (\$)		2000.00

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TOTAL FIRM MILESTONE 1 PRICE (\$) (including labour, materials, other costs, travel and cost incurred, excluding applicable taxes)	72000.00
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Milestone 2

Milestone 2 requires the following:

Progress reporting (Milestone 2) to be completed and delivered in accordance with the Statement of Work (SOW).

Total Performance Period (in weeks) 12 (weeks)				
Work Activities & Risks and Mitigation				
Activity	Description of Deliverable	Estimated Level of Effort (Weeks)	Risk(s): Description, probability and impact	Risk mitigation strategy(ies)
Implement the hardware enforcement datapath from the Milestone 1 architecture: RTL for the RISC-V core with the custom envelope-parse, label-match and disposition instructions; verification against the cycle-level simulation; and an FPGA prototype demonstrated on live bus traffic in both target profiles (server offload; sensor-adjacent for cameras and uncrewed platforms). Owned by the IC-design engineer.	The FPGA-prototyped enforcement datapath running against live bus traffic, with verification results and a measured comparison against the software gate — the silicon-ready proof for the Component 1b tape-out path.	3.00	RTL verification is the dominant hardware cost and could compress the FPGA demo (medium probability). Impact: the prototype demonstrates fewer policy classes in hardware than planned.	The hybrid escalation design makes partial hardware coverage safe — anything not proven in hardware runs through the software FCE, so the project deliverables never depend on hardware completeness. Verification reuses the recorded-traffic test vectors from Milestone 1 simulation.

Build the FCE enforcement engine at the broker chokepoint: the synchronous policy-evaluation hook from the Milestone 1 design, rendering a permit, restrict, downgrade or segregate disposition for every element before any cross-domain merge, with live policy hot-reload (a policy-set swap mid-session with enforcement continuing uninterrupted).	The working enforcement engine at the chokepoint: per-element dispositions on live AIS + SAR traffic under the synthetic Protected-B policy set, with demonstrated mid-session policy hot-reload.	2.00	The enforcement engine is the project's central new build (medium probability of design-to-build friction). Impact: schedule pressure on the demonstration activities later in the milestone.	The engine lands on a built, benchmarked substrate the applicant owns end-to-end — the hook is additive at one well-understood point, not a rebuild. The Milestone 1 design specification, fail-safe semantics and pre-compiled policy lookup bound the build risk, and demo scope is cut before the timeline extends.
Implement the enforcement-action record type in the CRC32 append-only write-ahead log — rule applied, action taken, resulting disposition, journalled immutably for every decision including operator overrides — and deliver the exportable ingestion-to-output lineage format modelled on W3C PROV, reconstructable by replay for	Immutable enforcement-action audit records in the WAL for every disposition and override, plus the exportable W3C-PROV-modelled lineage format with a demonstrated ingestion-to-output reconstruction.	2.00	Lineage export could grow large or slow on long scenarios (low probability). Impact: slower export runs; no effect on in-path enforcement, which never waits on export.	Export is an offline replay over the WAL, decoupled from the enforcement path by design; scenario-scoped exports and the existing bit-exact replay machinery keep it bounded.

compliance review, forensic analysis or accreditation.				
Deliver the evaluator-operable live demonstration at planetar.ca: the Technical Authority can drive the running system during evaluation — change a policy and watch enforcement hot-reload, and click any disposition back through its governing rule and the element's provenance, with override as a logged, attributable action. Built against the operator-accessibility and English/French readiness checklist (keyboard navigation, screen-reader compatibility, colour-vision-safe disposition encoding), delivered as a project artefact.	The live, evaluator-operable planetar.ca FCE demonstration (two modalities, synthetic Protected-B markings) plus the documented accessibility and bilingual-readiness checklist.	2.00	Demonstrator scope could overrun (medium probability). Impact: less time for the evaluation and benchmark activity at the end of the milestone.	The analyst shell, live feeds and provenance drill-down already exist (planetar-ui); this activity adds the disposition surface and policy controls. Non-essential viewer polish is cut before the schedule slips.
Measure and evaluate: (a) FCE-on enforcement overhead				

against the no-enforcement baseline on the shared-memory path, reported conservatively (bracketed measured runs, no post-tuning figures) with the FCE-on compute/memory envelope for edge deployment; (b) policy-decision correctness on the labelled synthetic Protected-B set; (c) lineage-export fidelity against ground truth; (d) the disposition bias-audit protocol — enforcement decisions stratified by source, modality and region to surface uneven handling (the GBA-Plus artefact).	The measurement report: enforcement overhead vs baseline, edge compute/memory envelope, policy-decision correctness, lineage-export fidelity, and the stratified disposition bias audit.	3.00	Measured enforcement overhead could exceed expectations for complex policy sets (low-medium probability). Impact: the reported overhead bracket is wider than hoped; results remain honest and useful.	The same conservative discipline as the existing benchmark: report the measured bracket whatever it is, stratify by policy complexity so the common case is visible, and use the pre-compiled-lookup design from Milestone 1 to keep the common path fast.
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Milestone 2 - Labour			
Category	Labour (h)	Rate (\$/h)	Total (\$)
IC-Design Engineer (hardware enforcement datapath)	230	140.00	32200.00
Principal Investigator (PhD, Computer Science / Machine Learning)	500	140.00	70000.00
Total Labour Cost (\$)			102200.00

Milestone 2 - Materials			
Description	Quantity	Unit Cost (\$)	Total (\$)
FPGA development board and prototyping sundries (RISC-V-capable, for the hardware enforcement-datapath prototype)	1	3800.00	3800.00
Total Materials Cost (\$)			3800.00

Milestone 2 - Other Costs		
Other cost	Description	Cost (\$)
Cloud compute	Cloud compute for enforcement-overhead benchmark runs, policy-correctness and lineage-fidelity evaluation, Sentinel-1 processing, and hosting the live evaluator-operable planetar.ca demonstrator.	2000.00
Software / tools	Development and operations tooling: continuous-integration runner minutes, observability/monitoring, and ancillary paid developer services. RTL flow on open-source toolchains.	500.00
Datasets / licences	Reserve for paid data access if a public source proves insufficient; core demonstration data (AIS, Sentinel-1) are public and free.	500.00
Total Labour Cost (\$)		3000.00

TOTAL FIRM MILESTONE 2 PRICE (\$) (including labour, materials, other costs, travel and cost incurred, excluding applicable taxes)	109000.00
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Location of Work

The Offeror will perform the work, including travel and living, in the following location(s)
(Maximum 256 characters per table entry)

Note: If none, Canada will leave this article blank or delete it.

City	Province / Territory	Country
Victoria	British Columbia	Canada

Language of Work

The contract will require work to be performed in either one or both of the [Official Languages of Canada](#). The Offeror is to indicate its capability:

~ *English*

Glossary

The following terms are used in this Proposal and have the following definitions:

If acronyms and/or uncommon terms are utilized, the Offeror is to add and define them below.

Note: If none, Canada will leave this article blank or delete it.

Acronyms and/or uncommon terms	Definitions
WAL (write-ahead log)	An append-only, integrity-protected log recording every event so state can be replayed exactly — the FCE's audit-and-lineage spine.
W3C PROV	The W3C standard data model for provenance and lineage; the interoperable target for the FCE's exportable lineage.
ABAC / XACML	Attribute-Based Access Control, and its OASIS policy-language standard; the incumbent access-decision approach the FCE contrasts with.
ODRL (Open Digital Rights Language)	A W3C standards-track machine-readable policy language; the FCE's policy substrate.
FCE (Fusion Compliance Engine)	A modular, AI-enabled component that automatically enforces classification and release-authority policy in real time during multi-sensor fusion.
FPGA (Field-Programmable Gate Array)	Reconfigurable hardware used to prototype and validate a digital circuit design before committing it to fixed silicon; the project demonstrates the hardware enforcement datapath on an FPGA.
RISC-V	An open, royalty-free processor instruction-set architecture; the hardware enforcement datapath is a RISC-V core extended with custom instructions for envelope parse, label match, and disposition.
Tape-out / multi-project wafer (MPW)	Committing a verified chip design to fabrication; an MPW run shares one mask set across many small projects, reducing test-chip cost (the Component 1b silicon path).
Disposition	The FCE's decision on a data element — permit, restrict, downgrade, or segregate.
STANAG 4774 / 4778	NATO standards for confidentiality metadata label syntax (4774) and for binding that label to the data object (4778).
Provenance / lineage	The recorded chain of which inputs and rules produced a given output, enabling audit, forensic review and accreditation.

Reference Documents

If the work details include references to documents critical for understanding the basis of the work, the Offeror is to fully describe these documents here.

Note: If none, Canada will leave this article blank or delete it.

Title	Author	Publication Date	Relevance to the Work
[S2] NATO STANAG 4778 (ADatP-4778, Edition A Version 1) — Metadata Binding Mechanism	NATO Standardization Office	2018-10-26	Binds a confidentiality label (STANAG 4774) to the data object across its lifecycle — the standards basis for provenance that travels with each element, and the coalition-interoperability target.
[P4] NIST Special Publication 800-162 — Guide to Attribute Based Access Control (ABAC) Definition and Considerations	NIST (National Institute of Standards and Technology)	2019-08-02	The reference formulation of attribute-based decisioning; the FCE's decision over each element's classification, domain, source, modality and releasability is an ABAC formulation applied in-path rather than at an access gate.
[P3] eXtensible Access Control Markup Language (XACML) Version 3.0, OASIS Standard	OASIS	2013-01-22	The incumbent attribute-based access-decision standard the FCE contrasts with — XACML decides access at a policy decision point; the FCE renders per-element dispositions in-path during fusion.

[P1] ODRL Information Model 2.2 (and ODRL Vocabulary and Expression 2.2)	W3C (World Wide Web Consortium) — W3C Recommendations	2018-02-15	Machine-readable, standards-track rights/policy language.
US Patent 11,442,952 B2 — User interface for commerce architecture	Steven R. Ness, one of 11 named inventors (assignee: Salesforce, Inc. — named-inventor credit only, not the assignee or owner; issued from App. 16/264,391)	2022-09-13	Canonical-data-model matching / identity-reconciliation prior art the applicant contributed to — secondary background credential for the provenance-and-lineage substrate, distinct from the project's net-new FCE foreground IP.
US Patent 10,936,582 B2 — Integrated entity view across distributed systems	Steven R. Ness, one of 19 named inventors (assignee: Salesforce, Inc. — named-inventor credit only, not the assignee or owner)	2021-03-02	The integrated-entity-view / entity-resolution architecture the applicant contributed to; planetar-ontology (the FCE's per-edge lineage graph tracking provenance from ingestion through fused output) is a substantial development beyond it.
The Orchive: A System for Semi-Automatic Annotation and Analysis of a Large Collection of Bioacoustic Recordings	Steven R. Ness (PhD thesis, University of Victoria, Dept. of Computer Science; supervisor George Tzanetakis)	2013-12-23	The applicant's doctoral work — archive-scale machine learning.

[A1] Automatic Event Detection for Long-Term Monitoring of Hydrophone Data	F. Sattar, P. Driessen, G. Tzanetakis, S. R. Ness, W. Page	2011-08-23	Peer-reviewed event detection on Ocean Networks Canada operational hydrophone data (IEEE PacRim 2011) — precedent for reliable analytics on continuous real-world sensor streams.
[S1] NATO STANAG 4774 (ADatP-4774, Edition A Version 1) — Confidentiality Metadata Label Syntax	NATO Standardization Office	2017-12-20	The NATO standard for a machine-readable confidentiality label (policy id, classification level, releasability) — the marking syntax the FCE's classification fields are designed to.
[A2] ORCA-SLANG: An Automatic Multi-Stage Semi-Supervised Deep Learning Framework for Large-Scale Killer Whale Call Type Identification	C. Bergler, M. Schmitt, A. Maier, H. Symonds, P. Spong, S. R. Ness, G. Tzanetakis, E. Noeth	2021-08-30	Applicant's co-authored, peer-reviewed semi-supervised deep learning on continuous, scarcely-labelled sensor streams (Interspeech 2021) — evidence of the capability to deliver reliable ML over noisy real-world data.